

Jathurshan Rasathurai

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• <https://jathurshan-r.github.io>

SUMMARY

Cybersecurity undergraduate with hands-on exposure to offensive security, threat analysis, and system defense through academic and self-built projects, TryHackMe, and Hack The Box. Strong interest in penetration testing, vulnerability exploitation, and attack simulation, with foundational knowledge of security monitoring, access control, and incident analysis. Actively seeking entry-level roles or internships in cybersecurity, including Red Team, SOC Analyst, and Cybersecurity Intern positions.

PROJECTS

Multi-threaded TCP Port Scanner with Banner Grabbing *Python | Socket Programming | Concurrent Programming (Threading) | argparse*

- Built a multi-threaded TCP port scanner using raw sockets, with no reliance on Nmap or third-party libraries.
- Added protocol-aware banner grabbing for SSH, FTP, SMTP, HTTP, and HTTPS.
- Scanned ~10,000 ports in under 3 seconds using a bounded ThreadPoolExecutor worker pool.
- Generated structured JSON and HTML reports with timestamped output.
- Packaged as a modular, MIT-licensed Python package on GitHub.

Repository: <https://github.com/JATHURSHAN-R/py-port-scanner>

Northwind Logistics Breach — Multi-Stage CTF Play Box *Python | Flask | Docker Compose | CTFd | Scapy | Nginx | Ubuntu | steghide | exiftool*

- Designed a six-stage CTF spanning OSINT, steganography, web security, cryptography, networking, and Linux security.
- Containerized all challenges with Docker Compose, including a vulnerable Flask app (SQLi, IDOR) and an SSH target with a SUID privilege-escalation path.
- Embedded flags using real steganography (steghide), PDF metadata tricks (exiftool), and a custom Scapy packet capture for Wireshark analysis.
- Integrated CTFd for scoring and a live scoreboard; tested end-to-end on a Kali VM.
- Hardened containers with cap_drop: ALL, read-only filesystems, and network segmentation.

Repository: <https://github.com/JATHURSHAN-R/northwind-ctf>

Secure DevSecOps CI/CD Pipeline — Vulnerability Remediation & Automated Security Gates *PHP | MariaDB | Docker Compose | GitHub Actions | Semgrep | Trivy | Gitleaks | HashiCorp Vault*

- Ran STRIDE threat modelling on a two-tier Dockerised app, mapping six threats to specific controls.
- Found and fixed four vulnerabilities (SQL injection, weak session IDs, file upload RCE, stored XSS), verified by re-exploiting the patched code.
- Wrote custom Semgrep rules, cutting SAST findings from 13 to 6.
- Built a GitHub Actions pipeline with four security gates (SAST, SCA, secrets scanning, container scanning); Trivy blocks build on CRITICAL/HIGH CVEs.
- Investigated the March 2026 trivy-action supply-chain attack and pinned CI/CD dependencies to commit hashes in response.
- Managed secrets via GitHub Actions and HashiCorp Vault, removing hardcoded credentials.

Repository: <https://github.com/JATHURSHAN-R/DVWA>

Machine Learning-Based Personalized Medical Diet Recommendation System *Python | scikit-learn | Pandas | Jupyter Notebook*

- Built a machine learning model to generate personalised diet plans from medical and nutritional data.
- Compared KNN, Decision Tree, Random Forest, and hybrid models for recommendation accuracy.
- Considered data privacy, bias, and patient safety in the model design.

Repository: <https://github.com/JATHURSHAN-R/Diet-Recommendation-ML>

EDUCATION

BSc (Hons) in Information Technology, specializing in Cyber Security

Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka

CERTIFICATIONS

- eJPT (INE Junior Penetration Tester) 2026 — Practical penetration testing across networks, web applications, and host systems; builds attacker-perspective knowledge that strengthens detection and triage.

- SOC Level 1 Learning Path — TryHackMe, 2026 — Security Operations Center fundamentals covering log analysis, SIEM tooling, threat intelligence, phishing analysis, and incident response workflows.
- ISC2 Certified in Cybersecurity (CC), 2026 — Certified professional with expertise in cybersecurity fundamentals, risk management, and compliance.
- TryHackMe – Advent of Cyber 2025 — Hands-on cybersecurity challenge covering real-world attack and defense scenarios.
- Hack The Box Academy – Linux Fundamentals — Foundational Linux skills including file systems, permissions, processes, and user management.
- Hack The Box Academy – Web Requests — Core web communication concepts such as HTTP methods, headers, cookies, and authentication mechanisms.

SKILLS

Cybersecurity Tools: Nmap, Wireshark, Metasploit, Burp Suite, Semgrep, Trivy, Gitleaks, steghide, exiftool, Scapy

Security Practices: Penetration Testing, Vulnerability Assessment, Threat Modeling (STRIDE), SAST, OWASP Top 10, Incident Response, SIEM

Programming Languages: Python, Java, SQL, C++, C

Network & Systems: Socket Programming, Concurrent Programming (Threading), TCP/IP, Linux (Ubuntu, Kali), Docker, Docker Compose

DevOps & CI/CD: GitHub Actions, CI/CD Pipeline Security, HashiCorp Vault

Web Technologies: HTML, CSS, JavaScript, Java Servlets, Spring Boot, Flask

Databases: MySQL, SQLite

Machine Learning: scikit-learn, Pandas, Model Evaluation

Tools: Git, GitHub, IntelliJ IDEA, VS Code, Jupyter Notebook, VirtualBox, CTFd

Languages: English, Tamil

REFEREE

Shashika Lokuliyana

Senior Lecturer | Sri Lanka Institute of Information Technology

M.Sc. (IT) SLIIT, B.Sc. (Sp.) Hons in IT (Spec. in CSN), SMIEEE, MCSSL, MISOC

Cisco Academy Instructor: CCNA, CyberOps, DevOps • Chair — IEEE WIE AGSL 2026 • Publicity Co-Chair — ICAC 2026

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